

SOC Internship Day 30

Cyber Deception Technology

Objective

Learn how Security Operations Center (SOC) teams use deception technologies to detect attackers within networks.

What is Cyber Deception?

Cyber Deception is a cybersecurity strategy that uses fake systems, services, credentials, and data to lure attackers and detect malicious activity. Since legitimate users should not interact with these decoys, any interaction is treated as highly suspicious.

Importance of Deception Technology

Cyber deception helps organizations:

- Detect attackers at an early stage
- Identify insider threats
- Improve threat intelligence collection
- Reduce attacker movement across networks

Common Deception Technologies

1. Honeypots

Fake systems designed to attract attackers and monitor their actions.

2. Honeytokens

Fake files, credentials, or records that trigger alerts when accessed.

3. Fake Credentials

Decoy usernames and passwords placed within systems to detect unauthorized use.

4. Decoy Servers

Non-production servers that imitate valuable targets.

5. Fake Databases

Databases containing fabricated information used to monitor attacker behavior.

How Honeypots Work

A honeypot appears vulnerable and attractive to attackers. When an attacker interacts with it:

1. An alert is generated.
2. Activity is logged.
3. The SOC team begins an investigation.

Example Investigation

Alert

Connection to a decoy server detected.

Findings

- Unauthorized login attempts
- Suspicious PowerShell execution

Analysis

The activity may indicate attacker reconnaissance or an attempt to gain access to critical systems.

Detection Techniques

SOC teams use various methods to identify malicious activity:

- Monitoring honeypot access
- Detecting fake credential usage
- Analyzing attacker behavior
- Tracking lateral movement across the network

SOC Workflow in Deception Technology

Attacker Activity

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Deception Alert

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SOC Investigation

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Threat Containment

SOC Team Roles

L1 Analyst

Monitors deception alerts and identifies suspicious activity.

L2 Analyst

Investigates attacker actions and validates threats.

L3 Analyst

Enhances deception strategies and improves defensive measures.

Key Takeaways

- Cyber deception helps detect threats that traditional security controls may miss.
- Honeypots and honeytokens provide early warning signs of attacker activity.
- Deception technology improves visibility into attacker techniques and behavior.
- SOC analysts use deception alerts to investigate and contain threats quickly.

Assignment Answers

1. Explain Cyber Deception

Cyber deception is a security technique that uses fake assets such as honeypots, decoy servers, and fake credentials to detect attackers. Any interaction with these assets is considered suspicious and helps security teams identify threats early.

2. Research Honeypots

A honeypot is a deliberately vulnerable system designed to attract attackers. It records attacker activities, techniques, and tools, helping organizations improve threat detection and incident response.

3. Analyze an Attacker Interaction Scenario

If an attacker connects to a decoy server and attempts unauthorized logins while running PowerShell commands, the SOC receives an alert. Analysts investigate the activity, determine the attacker's objectives, collect evidence, and take containment actions to prevent further compromise.